

Literature Status: The $\ell_2 \oplus \mathcal{T}^{(2)}$ UTAP Subquestion

Run `fa.banach_001`, lane 4

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Status

literature_already_answered, scoped subquestion only.

The source paper is F. Albiac, J. L. Ansorena, and P. Wojtaszczyk, *Uniqueness of unconditional basis of $H_p(\mathbb{T}) \oplus \ell_2$ and $H_p(\mathbb{T}) \oplus \mathcal{T}^{(2)}$ for $0 < p < 1$* , arXiv:2010.01501. In the final discussion after Theorem 4.7, it lists as one of the main remaining Bourgain–Casazza–Lindenstrauss–Tzafriri style cases whether

$$\ell_2 \oplus \mathcal{T}^{(2)}$$

has a UTAP unconditional basis.

The supporting paper is F. Albiac and J. L. Ansorena, *Uniqueness of unconditional basis of $\ell_2 \oplus \mathcal{T}^{(2)}$* , arXiv:2012.06783. This is a separate later paper and it explicitly knows the provenance: its introduction says it is a direct continuation of the 2020 work, explains that $\ell_2 \oplus \mathcal{T}^{(2)}$ lies outside the previous methods, and then says that the paper fills this gap by proving, in particular, that $\ell_2 \oplus \mathcal{T}^{(2)}$ has a UTAP unconditional basis. The abstract states the same conclusion.

Why This Answers The Source Subquestion

The source question is an exact yes/no UTAP question for the direct sum $\ell_2 \oplus \mathcal{T}^{(2)}$. The supporting paper’s title, abstract, and main application all identify the same direct sum and prove the affirmative UTAP conclusion. Therefore that subquestion is already answered in the literature.

Scope Limitations

This packet does not claim a solution of the broad question whether $X \oplus Y$ has a UTAP unconditional basis whenever both Banach spaces have one. It also does not settle the later Gowers-square question from arXiv:2603.07324 asking whether G^m has a unique unconditional basis for $m \geq 2$. The existing local attempt `attempts/2010.01501_gowers_square_vector_peak_reduction.md` remains the relevant no-promotion note for that route.

Evidence Checked

Local cheap indexes showed no prior solution packet for arXiv:2010.01501, but did show the Gowers-square failed attempt. The parsed local sources for arXiv:2010.01501, arXiv:2012.06783, and the later Gowers route source arXiv:2603.07324 were inspected.